

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
LAHONTAN REGION**

**BOARD ORDER NO. R6T-2002-0030
WDID NO. 6A290011000**

REVISED WASTE DISCHARGE REQUIREMENTS

FOR

**TAHOE-TRUCKEE SANITATION AGENCY
MARTIS VALLEY WASTEWATER TREATMENT PLANT
AND ASSOCIATED MAINTENANCE ACTIVITIES**

Nevada County

The California Regional Water Quality Control Board, Lahontan Region (Regional Board) finds:

1. Discharger

On January 10, 2002, the Tahoe-Truckee Sanitation Agency (TTSA) submitted a completed revised Report of Waste Discharge for the expansion of the Martis Valley Wastewater Treatment Plant. For the purpose of this Order, the Tahoe-Truckee Sanitation Agency (TTSA), as the operator, facility owner, and landowner, is referred to as the "Discharger".

2. Facility

For purposes of this Order, the land, buildings and equipment associated with the operations of the Martis Valley Wastewater Treatment Plant, and associated routine maintenance activities, are referred to as the "Facility".

3. Facility Location

The Facility is located at 13720 Joerger Drive, within the town limits of the Town of Truckee, Nevada County APN 49-010-20, as shown in Attachment "A", which is made a part of this Order. The Facility location is within Sections 7 and 12, T17N, R17E, MDB&M, which is within the Truckee River Hydrologic Unit. The Truckee River borders the northern side of the Facility and Martis Creek is located east of the Facility.

4. Order History

Regulation of the Facility began with the adoption of waste discharge requirements (WDRs) under Board Order No. 6-74-44, adopted April 25, 1974. The Board revised those WDRs under Board Order No. 6-77-27 adopted May 12, 1977. At the request of TTSA, the Board revised WDRs in 1981 by adopting Board Order No. 6-81-61, authorizing a 2.57 million gallons per day (MGD) maximum 7-day average flow capacity increase (from 4.83 to 7.4 MGD) upon completion of a proposed treatment facility expansion. The Board again revised

WDRs under Board Order No. 6-81-71 on September 17, 1981, authorizing the treatment facility capacity to be immediately increased from 4.83 to 5.83 MGD. The 1.0 MGD increase in capacity was part of the proposed 2.57 MGD capacity increase mentioned above, and occurred before the proposed treatment facility expansion was completed. The Board then updated the WDRs on February 19, 1987 by adopting Board Order No. 6-87-21, and again updated the WDRs on April 11, 1990 by adopting Board Order No. 6-90-27. The Board amended the WDRs on October 3, 1996, by adopting Board Order 6-90-27A1.

5. Reason for Action

The Regional Board is revising the WDRs to permit an increase in Facility capacity and change in treatment methods for nitrogen removal, and to modify the monitoring and reporting program in response to these changes. The revised WDRs also serve as an update as part of a statewide program to periodically review and update WDRs, and also to incorporate changes to conform to the revised Basin Plan.

6. TTSA as a Regional Entity

The TTSA is designated as the regional entity to transport, treat and dispose wastewater from the North Tahoe Public Utility District (NTPUD), Tahoe City Public Utility District (TCPUD), Alpine Springs County Water District (ASCWD), Squaw Valley Public Services District (SVPSD), Truckee Sanitary District (TSD) and the Truckee River Canyon area. Reference to member entities of the TTSA in the Lake Tahoe Basin specifically includes NTPUD and TCPUD which are subject to the requirements of the California-Nevada Interstate Water Compact, referenced in Finding No. 19. ASCWD, SVPSD, TSD and the Truckee River Canyon Area are not within the Lake Tahoe Basin; and therefore, are not subject to the California-Nevada Interstate Water Compact.

7. Description of Existing Facility

The wastewater treatment facility provides tertiary level treatment. The treatment processes consist of influent screening, grit removal, primary sedimentation, pure oxygen activated sludge, biological phosphorus removal, chemical treatment, mixed media filtration, ion exchange ammonia removal, and final chlorination. Organic sludge is digested anaerobically, dewatered and transported to a landfill. Waste chemical sludge is dewatered and also transported to a landfill. The Discharger is considering alternative disposal sites for sludge, which will be subject to Regional Board staff review and approval. Emergency storage of wastewater is provided at the former TSD wastewater treatment ponds.

8. Existing Facility Capacity

The TTSA is capable of transporting, treating and disposing of a maximum 7-day average municipal wastewater flow, during the summer months of 7.4 MGD.

9. Description of and Capacity of Proposed Expansion

The proposed expansion of the Facility will be capable of transporting, treating and disposing of a maximum 7-day average flow during the summer months of 9.6 MGD. The proposed expansion will include numerous replacement, upgraded, modified and additional components and units to provide additional capacity and improve treatment, as described as Alternative 3 in the *Draft Project Report* dated April 1999 and modified by the *Updated Project Report* dated January 2002. The proposed expansion will also replace the existing ion-exchange nitrogen removal system with a Biological Nitrogen Removal (BNR) system for the full 9.6 MGD capacity of the enlarged plant. The existing ion exchange process shall only be operated if there is a process upset during the startup of the new BNR process. Following a three-month period of successful operation of the new BNR process, the ion exchange process will only be operated as a standby unit for emergencies. Additional treatment provided by the Soil Aquifer Treatment (SAT) process will also be acknowledged. The expansion will also incorporate enlargement and improvements to the Truckee River Interceptor (TRI) and the emergency storage facilities at the former TSD Sewage Treatment Lagoons.

The proposed expansion also identifies additional improvements which will not be constructed immediately, but rather at a later date when needed. Among these improvements are an additional disposal field and a spray irrigation system, which will require subsequent approval by the Regional Board.

10. Point of Effluent Disposal

Plant effluent is discharged to subsurface disposal trenches, the boundaries of which are shown on Attachment "A", which is made a part of this Order. The disposal field is located in a portion of the SE/4, Section 7, T17N, R17E, MDB&M and are within the Truckee River Hydrologic Unit.

A second disposal field will be constructed when needed for additional disposal capacity. The location of the proposed second disposal field is also shown on Attachment "A". A pilot project operated in 1991 demonstrated the technical feasibility of disposing effluent by a spray irrigation system from April through November. The proposed location of a full-scale effluent spray irrigation system is also shown on Attachment "A". The Discharger does not intend to construct the full-scale effluent spray irrigation system until a later date when needed to meet effluent limitations. The proposed location of a full-scale spray irrigation system is also shown on Attachment "A". Other than small volumes used for plant irrigation, the three areas shown on Attachment "A" are the only designated disposal sites.

11. Site Hydrogeology

Soils investigations of the effluent disposal areas indicate that they are located over permeable glacial outwash (Tahoe outwash) deposits 70 to 100 feet thick. The sites are further underlain by the relatively impermeable clayey deposit of the Truckee Formation. Ground water elevations in both disposal areas are known to be at least 40 feet below the ground surface.

Hydrogeologic investigations, a mathematical simulation model, and bromide tracer studies indicate that plant effluent discharged to the subsurface disposal system will migrate from the disposal site toward the Truckee River and Martis Creek, a tributary of the Truckee River. The Truckee River and Martis Creek are both within a half mile of the disposal sites.

Additional studies have been conducted by TTSA to demonstrate and quantify the ability of the aquifer to remove nitrogen, phosphorus and bacteriological constituents in treated effluent.

12. Lahontan Basin Plan

The Regional Board adopted the *Water Quality Control Plan for the Lahontan Region* (Basin Plan) on March 31, 1995. This Order implements the Basin Plan, as amended.

13. Receiving Waters

The Facility discharges to ground waters of the Martis Valley Ground Water Basin. Studies indicate that plant effluent discharged to the subsurface disposal system will migrate from the disposal site toward the Truckee River and Martis Creek, which are within a half mile of the disposal site.

14. Beneficial Uses of Ground Water

The beneficial uses of ground waters of the Martis Valley Ground Water Basin, as set forth and defined in the Basin Plan, are:

- a. municipal and domestic water supply;
- b. agricultural supply; and
- c. freshwater replenishment.

15. Beneficial Uses of Surface Water

The beneficial uses of the Truckee River, as set forth and defined in the Basin Plan, are:

- a. municipal and domestic water supply;
- b. agricultural supply;
- c. industrial service supply;
- d. ground water recharge;
- e. freshwater replenishment;
- f. hydropower generation;
- g. water contact recreation;
- h. non-contact water recreation;
- i. commercial and sport fishing;
- j. cold freshwater habitat;
- k. wildlife habitat;
- l. rare, threatened, or endangered species;

- m. migration of aquatic organisms; and
- n. spawning, reproduction, and development.

The beneficial uses of Martis Creek, as set forth and defined in the Basin Plan, are:

- a. municipal and domestic supply
- b. agricultural supply;
- c. ground water recharge;
- d. water contact recreation;
- e. non-contact water recreation;
- f. commercial and sport fishing;
- g. cold freshwater habitat;
- h. wildlife habitat;
- i. rare, threatened, or endangered species;
- j. migration of aquatic organisms; and
- k. spawning, reproduction, and development.

16. Control Measures for the Lake Tahoe Basin

The "Control Measures for the Lake Tahoe Basin" is incorporated within Chapter 5 of the Basin Plan. This Chapter incorporates control measures that were previously included in the *Lake Tahoe Basin Water Quality Plan*. Included in these control measures are prohibitions of the discharge or threatened discharge of solid or liquid waste, including earthen materials from any new subdivision, new development in stream environment zones, new development not in conformance with land capability, new development not offset by implementation of remedial erosion control measures, to ground or surface waters or to stream environment zones in the Lake Tahoe Basin. To implement those and other provisions necessary to protect the water quality of the Lake Tahoe Basin, the control measures require that the Regional Board, in establishing WDRs for sewerage agencies servicing the Lake Tahoe Basin, to include the following:

- a. Conditions shall be set in WDRs to prohibit the sewerage agencies from providing any connection serving new development which is not in accordance with the Plan.
- b. Conditions shall be set in WDRs to require the development of raw sewage overflow preventative maintenance and spill response programs.
- c. Conditions shall be set in WDRs to require the submission of annual reports providing updated estimates of available sewage treatment capacity within the respective sewerage systems.
- d. Conditions shall be set in WDRs to require the determination of which structures in the Lake Tahoe Basin are not connected to a sewerage collection, treatment and export system.

17. Martis Creek Watershed Phosphorus Study

The receiving water objectives for Martis Creek allow minimal assimilative capacity for additional phosphorus. The proposed project may cause phosphorus levels in Martis Creek to exceed receiving water limitations. The current mean of monthly means phosphorus concentration above TTSA's influence upon the creek is 0.05 mg/l, which is also the receiving water limitation for the entire creek. The TTSA projects that with or without the proposed expansion, by 2010 this concentration may be elevated to 0.06 mg/l. To identify phosphorus sources and assess control measures that could be imposed within the Martis Creek watershed to reduce phosphorus loading to the creek and provide additional assimilative capacity for future TTSA discharges, the TTSA will provide funding for and ensure the completion of a study to identify existing and future sources of phosphorus within the watershed, potential control measures that could be implemented, and a monitoring plan to evaluate the effect of such control measures upon Martis Creek throughout its watershed.

18. Consideration of Water Rights

Section 174 of the California Water Code states in part:

"It is also the intention of the Legislature to combine the water rights and the water pollution and water quality functions of state government to provide for consideration of water pollution and water quality, and availability of unappropriated water whenever applications for appropriation of water are granted or waste discharge requirements or water quality objectives are established."

19. California-Nevada Interstate Compact

The California-Nevada Interstate Water Compact concerning the waters of the Lake Tahoe, Truckee River, Carson River and Walker River Basins was approved by the Legislatures of California and Nevada in 1970 and 1971, respectively. The United States Congress has not ratified the Compact. However, the states of California and Nevada are using the Compact as a guideline for allocation of water between the two states in those watersheds.

20. California Environmental Quality Act Compliance

An Environmental Impact Report for the proposed expansion of the Facility was adopted by the TTSA on December 19, 2000 in accordance with the provisions of the California Environmental Quality Act (Public Resources Code, §21000 et seq.). The Regional Board has considered the CEQA document and subsequent addendum to that document prepared by the lead agency. The following significant effects of the proposed expansion were identified in the CEQA document:

- a. Potential temporary significant short-term impact (increase in dust and noise) on local residential land uses adjoining TTSA facility during construction. Pursuant to CEQA, a Mitigation Monitoring Plan (MMP) has been prepared, summarizing mitigation that will be implemented to bring this potentially significant impact to nonsignificant levels, and also describes the duration that the mitigation will be implemented for and

includes the responsible parties for ensuring the success of the mitigation. The contractor will be responsible for implementation of the mitigation, and TTSA will be responsible for monitoring. The Regional Board is not responsible for implementation and monitoring of mitigation measures associated with this potentially significant impact.

- b. Potential temporary significant short-term impact (increase in dust and noise near residential land adjoining construction areas) during TSD ponds and TRI modifications. Pursuant to CEQA, a Mitigation Monitoring Plan (MMP) has been prepared, summarizing mitigation that will be implemented to bring this potentially significant impact to nonsignificant levels. The MMP also describes the duration that the mitigation will be implemented for and includes the responsible parties for ensuring the success of the mitigation. The contractor will be responsible for implementation of the mitigation, and TTSA will be responsible for monitoring. The Regional Board is not responsible for implementation and monitoring of mitigation measures associated with this potentially significant impact.

21. Public Notification

The Regional Board has notified the Discharger and all known interested parties of its intent to adopt updated waste discharge requirements for the Facility.

22. Consideration of Public Comments

The Regional Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that the Discharger shall comply with the following:

I. REVIEW PROCEDURES FOR NEW PROJECTS

- A. The TTSA shall submit to the Regional Board staff all projects which meet any of the following four criteria:
 - 1. Require a building permit
 - 2. Require a grading permit
 - 3. Have a soil disturbance of more than 1,000 square feet
 - 4. Propose soil disturbance within a stream environment zone
- B. No projects submitted for review per Review Procedure A. above may commence prior to the Executive Officer or the Regional Board approving the measures and facilities proposed for siltation and erosion control.

- C. During any emergency where the public health or welfare is threatened, the TTSA is authorized to take corrective action and shall use Best Management Practices for control of siltation and erosion as the situation demands; Regional Board staff shall be notified as soon as practical.

II. DISCHARGE SPECIFICATIONS

A. Effluent Limitations

1. The discharge to waters of the State shall not contain trace elements, pollutants, contaminants, or combinations thereof, in concentrations which are toxic to humans or to aquatic or terrestrial plant or animal life.
2. Treated wastewater made available for percolation shall not contain concentrations of parameters in excess of the following limits:

<u>Constituent</u>	<u>Effluent Limitations</u>	
	<u>Monthly average^a</u>	<u>Maximum^b</u>
Suspended Solids	10 mg/l	20 mg/l
Turbidity	---	10 NTU
Total Phosphorus	0.8 mg/l	1.5 mg/l
Chemical Oxygen Demand	45 mg/l	60 mg/l

^aThe "monthly average" is the arithmetic mean of measurements made during a month.

^bThe "daily maximum" is the highest daily 24-hour composite measurement during the monitoring period.

3. All treated wastewater made available for percolation shall have a dissolved oxygen concentration greater than 0.5 mg/l.
4. Treated wastewater made available for percolation shall have a total trihalomethanes concentration of less than 50 ppb, measured as an arithmetic mean of all samples taken during a calendar year.
5. Treated wastewater made available for percolation shall have a 7-day mean of no more than 23 total coliform organisms, and shall have a mean of any two consecutive samples of no more than 240 total coliform organisms.

6. Effective immediately and continuing until completion of the treatment plant expansion, or until completion of BNR for the existing rated capacity if expansion does not occur, treated wastewater made available for percolation shall not contain concentrations in excess of the following limits:

<u>Constituent</u>	<u>Effluent Limitations</u>	
	<u>Annual average^c</u>	<u>Maximum^b</u>
Total Dissolved Solids	600 mg/l	---
Chloride	200 mg/l	---

^bThe "daily maximum" is the highest daily 24-hour composite measurement during the monitoring period.

^cThe "Annual Average" is the arithmetic mean of all measurements made during a calendar year.

7. Effective immediately and continuing until four years after the completion of the treatment plant expansion, or until four years after the completion of BNR for the existing rated capacity if expansion does not occur, treated wastewater made available for percolation shall not contain concentrations in excess of the following limits:

<u>Constituent</u>	<u>Effluent Limitations</u>	
	<u>Monthly average^a</u>	<u>Maximum^b</u>
Total Nitrogen (as N)	9 mg/l	12 mg/l

^aThe "monthly average" is the arithmetic mean of measurements made during a month.

^bThe "daily maximum" is the highest daily 24-hour composite measurement during the monitoring period.

The TTSA shall use all existing wastewater treatment facilities capable of reducing the monthly average total nitrogen concentration below 9.0 mg/l in all treated wastewater made available for percolation. If objectionable alterations in the species composition of any surface waters occur in the biomass and/or objectionable alterations in the species composition of any surface waters occur as a result of percolating wastewater effluent, the TTSA shall reduce effluent nitrogen concentrations below the Effluent Limitation of 9.0 mg/l. The reduction shall be made within 30 days after being notified by the Regional Board.

B. Flow Limitations

1. Effective immediately, and continuing until the completion of the treatment plant expansion, the following flow limitations of the facility shall be effective:
 - a. From June 21 through September 21 of any year, the flow of wastewater to the treatment and disposal facilities during any seven (7) consecutive days shall not exceed an arithmetic average of 7.4 MGD.
 - b. The maximum instantaneous flow rate of wastewater through the treatment facilities shall not exceed 13.0 MGD.
2. Immediately after the completion of the treatment plant expansion, the following flow limitations of the facility shall be effective:
 - a. From June 21 through September 21 of any year, the flow of wastewater to the treatment and disposal facilities during any seven (7) consecutive days shall not exceed an arithmetic average of 9.6 MGD.
 - b. The maximum instantaneous flow rate of wastewater through the treatment facilities shall not exceed 15.4 MGD.
3. The TTSA and its member entities in the Lake Tahoe Basin shall not issue sewer connection permits to a new development unless the Regional Board has determined that the new development is consistent with the Lake Tahoe Basin Water Quality Plan. A determination by the TRPA of consistency can be relied upon by the TTSA and its member entities in the Lake Tahoe Basin unless the Regional Board specifies in writing otherwise. TTSA and its member entities in the Lake Tahoe Basin shall notify the Regional Board of any such determination made by the TRPA before issuing a sewer connection permit.
4. TTSA shall submit annual reports providing updated estimates of available sewage collection export and treatment capacity within its system. TTSA shall also submit in the same report, updated estimates of available sewage collection and export capacity within the individual systems of its member entities. These reports shall be submitted to the Regional Board not later than April 1 of each year, providing the following information for the previous calendar year:
 - a. The effective capacity of each key element of the collection, treatment, export and disposal systems.
 - b. Current high flows.

- c. An allocation of capacity among: (1) current users; (2) projects for which connection permits have been issued; (3) capacity currently used or to be reserved for public agencies; (4) projects for which will-serve letters or similar commitments have been issued; and (5) available capacity, listed in terms for total flow and single family dwelling unit equivalents. Available capacity is determined as the differences between items 1-4 and the effective capacity of the most limiting component of the wastewater system.
- d. The number of additional connection permits or service commitments to be issued in the coming year, and the flow projected from these units.
- e. The number of subdivided vacant residential, commercial or public service lots within its boundary which are not located in subdivisions where onsite domestic wastewater disposal has been approved indefinitely by all appropriate agencies.
- f. Any proposed actions, including time schedules and financial plans, which will provide increases in effective capacity.

C. Receiving Water Limitations

- 1. The discharge shall not cause the following receiving water limitations for the Truckee River Hydrologic Unit to be exceeded:

<u>Constituent</u>	<u>Units</u>	<u>Truckee River below Martis Cr ^{1/}</u>	<u>Truckee River at Stateline ^{1/}</u>
Total Dissolved Solids	mg/l	80	75
Chloride	mg/l	10	8
Sulfate	mg/l	5	5
Total Iron	mg/l	0.29	0.30
Nitrate Nitrogen	mg/l as N	0.20	0.08
Total Kjeldahl Nitrogen	mg/l as N	0.20	0.32
Total Nitrogen	mg/l as N	0.40	0.40
Total Phosphorus	mg/l as P	0.05	0.05
Boron	mg/l	---	1.0 ^{2/}

^{1/}Arithmetic mean of monthly means

^{2/}Maximum limitation

2. The discharge shall not cause the following receiving water limitations for Martis Creek to be exceeded:

<u>Constituent</u>	<u>Units</u>	<u>Mean</u> ^{3/}
Total Dissolved Solids	mg/l	150
Chloride	mg/l	25
Sulfate	mg/l	8
Total Iron	mg/l	0.40
Nitrate Nitrogen	mg/l as N	1.00
Total Kjeldahl Nitrogen	mg/l as N	0.45
Total Nitrogen	mg/l as N	1.45
Total Phosphorus	mg/l as P	0.05

^{3/} Arithmetic mean of monthly means

3. Effective immediately, and continuing for the life of the project, ground water containing treated wastewater which was made available for percolation (as measured at Well 31) shall not contain concentrations of parameters in excess of the following limits prior to entering Martis Creek and/or the Truckee River:

<u>Constituent</u>	<u>Effluent Limitations</u>	
	<u>Monthly average</u> ^a	<u>Maximum</u> ^b
Chemical Oxygen Demand	15 mg/l	40 mg/l
Un-ionized Ammonia (as N)	---	0.20 mg/l
Total Phosphorus (as P)	0.3 mg/l ^c	---
Fecal Coliform Bacteria	---	2.2 MPN/100 ml ^d

^aThe "monthly average" is the arithmetic mean of measurements made during a month.

^bThe "daily maximum" is the highest daily 24-hour composite measurement during the monitoring period.

^cAnnual Average

^d2.2 - mean of 7-day average

4. Effective immediately, and continuing for the life of the project, ground water containing treated wastewater which was made available for percolation (as measured at Well 31) shall have a pH of not less than 6.5 units nor greater than 8.5 units.
5. Beginning four years after completion of the treatment plant expansion, or four years after completion of BNR for the existing rated capacity if expansion does not occur, ground water containing treated wastewater which was made available for percolation (as measured at Well 31) shall not contain concentrations of total

for percolation (as measured at Well 31) shall not contain concentrations of total nitrogen in excess of the following limits prior to entering Martis Creek and/or the Truckee River:

Constituent	Monthly Average ^a	Daily maximum ^b
Total Nitrogen (as N)		
May 1-October 31	2.0 ^e	---
Annual Average (Jan 1-December 31)	3.0 ^e	---

^aThe "monthly average" is the arithmetic mean of measurements made during a month.

^bThe "daily maximum" is the highest daily 24-hour composite measurement during the monitoring period.

^eAverage of monthly averages for monitoring period. Note that in addition to the concentration requirements, the discharge shall not exceed the mass loading as shown below.

6. Effective immediately, and continuing for the life of the project, ground water containing treated wastewater which was made available for percolation (as measured at Well 31) shall not exceed the following mass loading prior to entering Martis Creek and/or the Truckee River:

Constituent	Mass Loading Limitation (lbs/day)
Total Phosphorus (as P)	24 lbs/day (12-month average) ^f
Total Nitrogen (as N)	
May 1- October 31	128 lbs/day (6-month average) ^f
Yearly Average	204 lbs/day (12-month average) ^f

^fAverage of monthly averages for six month period or for calendar year

7. On or before completion of the treatment plant expansion, or completion of BNR for the existing rated capacity if expansion does not occur, treated wastewater which was made available for percolation (as measured at Well 31) shall not exceed the following mass loadings prior to entering Martis Creek and/or the Truckee River:

Constituent	Mass Loading Limitation (lbs/day)
Total Dissolved Solids	AAF ^g x 415 mg/l x 8.345 (annual average)
Chloride	AAF ^g x 115 mg/l x 8.345 (annual average)

^gAAF = average annual flow

8. Beginning with completion of the treatment plant expansion, or with the completion of BNR for the existing rated capacity if expansion does not occur, and continuing for the next four years, ground water containing treated wastewater which was made available for percolation (as measured at Well 31) shall not exceed the following mass loading prior to entering Martis Creek and/or the Truckee River:

Constituent	Mass Loading Limitation (lbs/day)
Total Dissolved Solids	$AAF^g \times (35 \times (9.6 - M7DADF^h) / 2.2 + 360) \times 8.345$ or $AAF^g \times 395 \text{ mg/l} \times 8.345$, whichever is less (annual average)
Chloride	$AAF^g \times (15 \times (9.6 - M7DADF^h) / 2.2 + 100) \times 8.345$ or $AAF^g \times 115 \text{ mg/l} \times 8.345$, whichever is less (annual average)

^gAAF = average annual flow

^hM7DADF = maximum 7-day average dry weather flow

9. Beginning four years after completion of the treatment plant expansion, or four years after completion of BNR for the existing rated capacity if expansion does not occur, ground water containing treated wastewater which was made available for percolation (as measured at Well 31) shall not exceed the following mass loadings prior to entering Martis Creek and/or the Truckee River:

Constituent	Mass Loading Limitation (lbs/day)
Total Dissolved Solids	$AAF^g \times 360 \text{ mg/l} \times 8.345$ or 24,514, whichever is less (annual average)
Chloride	$AAF^g \times 100 \text{ mg/l} \times 8.345$ or 6,809, whichever is less (annual average)

^gAAF = average annual flow

10. Effective immediately and continuing throughout the life of the project, the discharge of treated wastewater effluent to ground waters shall not cause a violation of the following water quality objectives for waters of the Martis Valley Ground Water Basin:

- a. Tastes and Odors – The taste and odor of ground waters shall not be altered.
- b. Bacteria – In ground waters designated as MUN, the median concentration of coliform organisms over any seven-day period shall be less than 1.1/100 milliliters.
- c. Radioactivity – Ground waters designated as MUN shall not contain concentrations of radionuclides in excess of the limits specified in Table 4 of Section 64443 (Radioactivity) of Title 22 of the California Code of Regulations which is incorporated by reference into the Basin Plan. This incorporation-by-reference in prospective including future changes to the incorporated provisions as the changes take effect.
- d. Chemical Constituents – Ground waters designated as MUN shall not contain concentrations of chemical constituents in excess of the maximum contaminant level or secondary maximum contamination level based upon drinking water standards specified in the following provisions of Title 22 of The California Code of Regulations which are incorporated by reference into the Basin Plan: Table 64431-A of Section 64431 (Inorganic Chemicals), Table 64431-B of Section 64431 (Flouride), Table 64444-A of Section 64444 (Organic Chemicals, Contaminant Levels-Consumer Acceptance Limits), and Table 64449-B of Section 64449 (Secondary Maximum Contaminant Levels-Ranges). This incorporation-by-reference is prospective including future changes to the incorporated provisions as the changes take effect.

Ground waters designated as AGR shall not contain concentrations of chemical constituents in amounts that adversely affect the water for beneficial uses (i.e., agricultural purposes).

Ground waters shall not contain concentrations of chemical constituents in amounts that adversely affect the water for beneficial uses.

11. The operation of the facility shall not cause a violation of the following water quality objectives for surface waters of the Truckee River Hydrologic Unit:
 - a. Turbidity – The turbidity shall not be raised above 3 Nephelometric Turbidity Units (NTU) mean of monthly means. (This objective is approximately equal to the State of Nevada standard of 5 NTU sample mean).

- b. Floating Material – Waters shall not contain floating material, including solids, liquids, foams and scum, in concentrations that cause nuisance or adversely affect the water for beneficial uses.

For natural high quality waters, the concentrations of floating material shall not be altered to the extent that such alterations are discernable at the 10 percent significance level.

- c. Suspended Materials – Waters shall not contain suspended material in concentrations that cause a nuisance or adversely affect the beneficial uses.

For natural high quality waters, the concentrations of suspended material shall not be altered to the extent that such alterations are discernable at the 10 percent significance level.

- d. Settleable Material – Waters shall not contain substances in concentrations that result in deposition of material that causes nuisance or that adversely affects the water for beneficial uses. For natural high quality waters, the concentrations of settleable material shall not be raised by more than 0.1 milliliter per liter.

- e. Color – The color shall not exceed an eight (8) Platinum Cobalt Unit mean of monthly means (approximately equivalent to the State of Nevada standard of a twelve (12) Platinum Cobalt Unit sample mean).

- f. Tastes and Odors – The taste and odor shall not be altered.

- g. Algal Growth Potential – The mean monthly algal growth potential shall not be altered to the extent that such alterations are discernible at the 10 percent significance level. This objective does not apply to Martis Creek; however, nuisance and pollution levels of algal growth shall not be discernible at these stations.

- h. Biostimulatory Substances – The concentrations of biostimulatory substances shall not be altered in an amount that could produce an increase in aquatic biomass to the extent that such increases in aquatic biomass are discernible at the 10 percent significance level. See pg. 3-9 of Basin Plan for additional language.

- i. Species Composition – The species composition of aquatic organisms shall not be altered to the extent that such alterations are discernible at the 10 percent significance level. See pg. 3-9 of Basin Plan for additional language.

- j. pH – Changes in normal ambient pH levels shall not exceed 0.5 pH units.
- k. Dissolved Oxygen – The dissolved oxygen concentrations shall not be depressed by more than 10 percent, below 80 percent saturation, or below 7.0 mg/l, whichever is more restrictive.
- l. Bacteria – Waters shall not contain concentrations of coliform organisms attributable to anthropogenic sources, including human and livestock waste. The fecal coliform concentration during any 30-day period shall not exceed a log mean of 20/100 ml, nor shall more than 10 percent of all samples collected during any 30-day period exceed 40/100 ml.
- m. Temperature – The natural receiving water temperature shall not be altered unless it can be demonstrated to the satisfaction of the Regional Board that such an alteration in temperature does not adversely affect the water for beneficial uses.
- n. Toxicity – All waters shall be maintained free of toxic substances in concentrations that are toxic to, or that produce detrimental physiological responses in human, plant, animal or aquatic life.
- o. Pesticides – Pesticide concentrations, individually or collectively, shall not exceed the lowest detectable levels, using the most recent detection procedures available. There shall not be a an increase in pesticide concentrations found in bottom sediments. There shall be no detectable increase in bioaccumulation of pesticides in aquatic life.

Waters designated as MUN shall not contain concentrations of pesticides or herbicides in excess of the limiting concentrations specified in Table 64444-A of Section 6444 (Organic Chemicals) of Title 22 of the California Code of Regulations which is incorporated by reference into the Basin Plan. This incorporation-by-reference is prospective including future changes to the incorporated provisions as the changes take effect.

Pesticides are defined here and in the Basin Plan to include insecticides, herbicides, rodenticides, fungicides, piscicides, and all other economic poisons. An economic poison is any substance intended to prevent, repel, destroy, or mitigate the damage from insects, rodents, predatory animals, bacteria, fungi or weeds capable of infesting or harming vegetation, humans or animals.

- p. Oil and Grease – Water shall not contain oils, greases, waxes or other materials in concentrations that result in a visible film or coating on

the surface of the water or on objects in the water, that cause nuisance, or that otherwise adversely affect the water for beneficial uses.

- q. Sediment – The suspended sediment load and suspended sediment discharge rate of surface waters shall not be altered in such a manner as to cause nuisance or adversely affect the water for beneficial uses.
- r. Radioactivity – Radionuclides shall not be present in concentrations which are deleterious to human, plant, animal, or aquatic life nor which result in the accumulation of radionuclides in the food web to an extent which presents a hazard to human, plant, animal, or aquatic life.

Waters designated as MUN shall not contain concentrations of radionuclides in excess of the limits specified in Table 4 of Section 64443 (Radioactivity) or Title 22 of the California Code of Regulations which is incorporated by reference into the Basin Plan. This incorporation-by-reference is prospective including future changes to the incorporated provisions as the changes take effect.

- s. Non-degradation of Aquatic Communities and Populations – All wetlands shall be free from substances attributable to wastewater or other discharges that produce adverse physiological responses in humans, animals, or plants; or which lead to the presence of undesirable or nuisance aquatic life. All wetlands shall be free from activities that would substantially impair the biological community as it naturally occurs due to physical, chemical and hydrological processes.
- t. Chlorine, Total Residual – For the protection of aquatic life, total chlorine residual shall not exceed either a median value of 0.002 mg/l or a maximum value of 0.003 mg/l. Median values shall be based on daily measurements taken within any six-month period.
- u. Chemical Constituents - Waters designated as MUN shall not contain concentrations of chemical constituents in excess of the maximum contaminant level or secondary maximum contamination level based upon drinking water standards specified in the following provisions of Title 22 of The California Code of Regulations which are incorporated by reference into the Basin Plan: Table 64431-A of Section 64431 (Inorganic Chemicals), Table 64431-B of Section 64431 (Flouride), Table 64444-A of Section 64444 (Organic Chemicals, Contaminant Levels-Consumer Acceptance Limits), and Table 64449-B of Section 64449 (Secondary Maximum Contaminant Levels-Ranges). This incorporation-by-reference is prospective

including future changes to the incorporated provisions as the changes take effect.

Waters designated as AGR shall not contain concentrations of chemical constituents in amounts that adversely affect the water for beneficial uses (i.e., agricultural purposes).

- v. Un-Ionized Ammonia - The neutral, un-ionized ammonia species (NH_3^0) is highly toxic to freshwater fish. The fraction of toxic NH_3^0 to total ammonia species ($\text{NH}_4^+ + \text{NH}_3^0$) is a function of temperature and pH. Tables 3-1 to 3-4 of the Basin Plan were derived from USEPA ammonia criteria for freshwater. Ammonia concentrations shall not exceed the values listed in these tables. For temperature and pH values not explicitly in these tables, the most conservative value neighboring the actual value may be used or criteria can be calculated from numerical formulas developed by the USEPA. Waters shall not contain concentrations of chemical constituents in amounts that adversely affect the water for beneficial uses.

D. Emergency Storage

1. The TTSA shall continue to provide emergency wastewater storage facilities capable of preventing treatment and disposal facility overloading or unauthorized discharges due to excessive flows or system breakdowns.
2. Emergency storage facilities shall have a capacity of at least 24 million gallons in addition to what is normally stored in the 15 million gallon emergency retention basin during routine treatment procedures.
3. Emergency storage facilities shall be sealed to prevent percolation of wastewater. The offsite ponds "A" and "B" have been lined with one foot of bentonite clay and shall be maintained as necessary to ensure the liner integrity.
4. All stored sewage shall be pumped to wastewater treatment and disposal facility.
5. The discharge of untreated or partially treated wastewater to emergency storage facilities is prohibited, except when any of the following conditions occur:
 - a. Loss of electrical power at the wastewater treatment facility.
 - b. Major equipment failure at the wastewater treatment facility.
 - c. Wastewater treatment process upset.

- d. Excessive infiltration/inflow into sewage facilities.
 - e. Any other emergency that could threaten the public health.
 - f. Implementing collection system, treatment plant and/or disposal system maintenance programs.
6. When additional emergency storage is determined to be necessary by the discharger, improvements shall be made to the offsite ponds to increase their storage capacity.

E. Full-Scale Effluent Spraying System

The discharger does not intend to operate a full-scale effluent spray irrigation system at this time. If the system is to be in operation, operations shall comply with the following requirements:

1. Before commencing operation of the effluent spray irrigation system, the Discharger shall install ground water monitoring wells and collect at least twelve months of monitoring data to characterize pre-project quality and water levels. The monitoring wells shall later be incorporated into the compliance monitoring program for the effluent spraying system.
2. At least 120 days prior to construction of the effluent spray irrigation system, the Discharger shall submit to Regional Board staff final plans, a program for initial start-up of the system and proposed modifications to the compliance monitoring program. The Executive Officer must approve these plans prior to construction of the system. The Monitoring program will be modified at that time.
3. Effluent shall not drift on to any access road.
4. Surface flow of effluent shall not migrate beyond the Discharger's property.
5. This office shall be notified at least 24 hours in advance of the beginning of each spraying season.
6. The effluent spraying system shall not result in a pollution or nuisance in either Martis Creek or the Truckee River.

F. Pretreatment of Industrial Wastewaters

1. The Discharger shall perform pretreatment functions, as described in 40 CFR Part 403, to include the following:
 - a. Implement the necessary legal authorities as provided for in 40 CFR 403.8 (f)(1).

- b. Establish a waste hauler permit system that regulates waste haulers discharging to the TTSA treatment plant, to be approved by the Executive Officer.
 - c. Develop a local pretreatment program, according to 40 CFR 403.5, to include the following minimum requirements:
 - (1) Conduct an industrial waste survey to identify all industrial dischargers that might be subject to the pretreatment program.
 - (2) Determine the character and volume of pollutants contributed to the TTSA facility by these industries.
 - (3) Conduct a technical evaluation to determine the maximum allowance treatment plant headworks (influent) loading for at least cadmium, chromium, copper, lead, nickel, and zinc.
 - (4) Identify any additional pollutants of concern.
 - (5) Implement a system to assure these loadings will not be exceeded.
 - d. Perform ongoing industrial inspections and monitoring as necessary to ensure compliance with any applicable pretreatment regulations.
2. The Discharger shall submit annually a report to the Regional Board describing the Discharger's pretreatment activities over the previous twelve months. In the event that the Discharger is not in compliance with any conditions or requirements of this Board Order, then the Discharger shall also include the reasons for non-compliance and state how and when the Discharger shall comply with such conditions and requirements. This annual report is due on July 1 of each year and shall contain, but not be limited to, the attached Appendix "A" titled, "Requirements for Pretreatment Annual Report", which becomes a part of this Order.

G. Best Management Practices

1. Prior to any disturbance of existing soil conditions, the Discharger shall install temporary erosion control facilities to prevent transport of eroded earthen materials and other wastes off the property.
2. Vehicle use shall be prevented in unpaved areas not subject to construction.
3. There shall be no significant modification of existing drainage ways or existing stream channel geometry except for the purpose of stabilization or enhancement of water quality improvement effects. All modifications of the

bed, channel, or bank of a stream require a prior written agreement with the California Department of Fish and Game.

4. All soil disturbance activities shall cease and temporary erosion control measures immediately installed if adverse weather conditions threaten the transport of disturbed soils from the project site.
5. All disturbed areas shall be adequately restabilized or revegetated. Revegetated areas shall be continually maintained until vegetation becomes established.
6. Stormwater runoff collection, pretreatment, and/or infiltration disposal facilities shall be designed, installed, and maintained to preclude a discharge of stormwater runoff for at least a 20-year 1-hour design storm (approximately 0.75" of rainfall) from all impervious surfaces. If site conditions do not allow for adequate on-site disposal, all site runoff must be treated to meet Effluent Limitations and the Receiving Water Limitations.
7. Stormwater runoff in excess of the design storm shall only be discharged to a storm drain or stabilized drainage, and must meet the Effluent Limitations
8. Surface flows from the project site shall be controlled so as to not cause downstream erosion at any point.
9. Stormwater runoff handling and disposal facilities shall be cleaned and renovated annually.
10. All disturbed soils and surplus waste earthen materials shall be removed from the project site and deposited only at a legal point of disposal, or restabilized on-site in accordance with erosion control plans previously reviewed by the Executive Officer.
11. At no time shall waste earthen materials be placed in surface water drainage courses, or in such a manner as to allow the discharge of such matter to adjacent undisturbed land or to any surface water drainage course.
12. All loose piles of soil, silt, clay, sand, debris or other earthen material shall be protected in a reasonable manner to prevent any discharge to waters of the state.
13. Any dewatering of trenches shall be done in a manner so as to eliminate the discharge of soil, silt, clay, sand or other waste earthen materials from the site to nearby surface waters.
14. Any damage or break in existing water or sewer lines shall be repaired as soon as possible and measures must be implemented to prevent erosion or sedimentation into any drainage way.

15. Fresh concrete or grout shall not be allowed to contact or enter surface waters.
16. The Discharger shall immediately clean up and transport to a legal site any spilled petroleum products to the maximum extent practicable.

H. General Requirements and Prohibitions

1. There shall be no discharge, bypass or diversion of raw or partially treated sewage, sewage sludge, grease, or oils from the transport, storage, treatment or disposal facilities to adjacent land areas or surface waters.
2. The discharge of wastewater except to the designated disposal sites is prohibited.
3. All facilities used for transport, storage, treatment, or disposal of waste shall be adequately protected against overflow, washout or inundation from a storm or flood having a recurrence interval of once in 100 years.
4. All waste organic and chemical sludges shall only be discharged at a legal point of disposal.
5. Where any numeric or narrative water quality objective contained in the Basin Plan is already being violated, the discharge of waste which causes further degradation or pollution is prohibited.
6. The surfacing of wastewater effluent at the designated subsurface disposal site, or within a 50-foot wide zone surrounding the designated subsurface disposal site, is prohibited. This prohibition does not apply to the surfacing of wastewater effluent encountered outside the 50-foot wide zone surrounding the designated subsurface disposal site. This prohibition also does not apply to maintenance activities authorized by this Board Order that are located on the designated subsurface disposal site or within the 50-foot wide zone surrounding the designated subsurface disposal site. All other activities proposed within the designated subsurface disposal site or within the 50-foot wide zone surrounding the designated subsurface disposal site, which may encounter wastewater effluent, shall be submitted for review by Regional Board staff and approval by the Executive Officer.
7. Neither the treatment nor the discharge shall cause a pollution or nuisance as defined in Section 13050 of the California Water Code, or a threatened pollution.

III. PROVISIONS

A. Recission of Board Order No. 6-90-27

Board Order No. 6-90-27 and Board Order 6-90-27A1 are hereby rescinded.

B. Monitoring and Reporting

Pursuant to Section 13267(b) of the California Water Code, the Discharger shall comply with Monitoring and Reporting Program No. 2002-(TENTATIVE).

C. Standard Provisions

The Discharger must comply with the "Standard Provisions for Waste Discharge Requirements", included in Attachment "B", which is made part of this order.

D. Right to Revise Waste Discharge Requirements

In accordance with Section 13263(e) of the California Water Code, the Regional Board reserves the right to review and revise all or any portion of these waste discharge requirements. Such action may be initiated on the Regional Board's own motion or in response to an application by any person affected by the discharge, for good cause, including the possibility that land uses in the area may change.

E. Wastewater Treatment Plant Operator Certificate

The Discharger's wastewater treatment plant shall be supervised by persons possessing a wastewater treatment plant operator certificate of appropriate grade pursuant to Chapter 3, Subchapter 14, Title 23, California Code of Regulations.

F. Addition of Biological Nitrogen Removal

TTSA shall initiate construction of full Biological Nitrogen Removal (BNR) for the existing rated capacity of 7.4 mgd by June 15, 2004, if the plant expansion is delayed or abandoned; provided that a total grant of \$11.6 million has been committed and made available to TTSA by the State of California for this purpose. The BNR process shall be designed for maximum practicable nitrogen reduction, independent of additional removals that can be achieved in the soil aquifer treatment (SAT) system. Operational measures shall be employed to maximize the overall performance of the BNR and SAT systems in concert with one another, to minimize nitrogen discharged to the Truckee River.

G. Martis Creek Watershed Phosphorus Study

Prior to **January 1, 2003**, the Discharger shall submit a workplan for a study to identify existing and future sources of phosphorus within the Martis Creek watershed, potential control measures that could be implemented, and a monitoring plan to

evaluate the effect of such control measures upon Martis Creek throughout its watershed. The workplan shall describe the study and include a time schedule for its completion by **June 30, 2004**. Though the study will involve and require the participation of other public agencies and private entities, the Discharger shall provide funding for and ensure the completion of the study.

H. Sewage Overflow Preventative Maintenance and Spill Response Programs

Member entities of the TTSA in the Lake Tahoe Basin shall develop sewage overflow preventative maintenance and spill response programs as specified in their waste discharge requirements. The TTSA shall maintain and update as necessary a sewage overflow preventative maintenance and spill response program for interceptor sewerlines which it maintains. The program for the interceptor shall be updated, and resubmitted to the Board for review and approval, at least once every three years, with the next report due no later than January 1, 2003.

I. Toxic Effluent Standards and Prohibitions

If a toxic effluent standard or prohibition (including any schedule of compliance specified in such effluent standard or prohibition) is established under Section 307(a) of the Federal Water Pollution Control Act or amendments thereto, for toxic pollutants contained in wastewater, the Board may revise or modify this Order in accordance with such toxic pollutant guidelines and so notify the discharger.

J. Required Connection to Sewer System in Lake Tahoe Basin

Member entities of the TTSA in the Lake Tahoe Basin shall continue to require the connection to the sewer system of any building from which waste is discharged, in accordance with Section 13950 of the California Water Code and as specified in their respective waste discharge requirements.

I, Harold J. Singer, Executive Officer, do hereby certify that the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Lahontan Region, on May 9, 2002.


HAROLD J. SINGER
EXECUTIVE OFFICER

Attachment: A. Location Map
B. Standard Provisions for Waste Discharge Requirements

ATTACHMENT "B"

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
LAHONTAN REGION

STANDARD PROVISIONS
FOR WASTE DISCHARGE REQUIREMENTS

1. Inspection and Entry

The discharger shall permit Regional Board staff:

- a. to enter upon premises in which an effluent source is located or in which any required records are kept;
- b. to copy any records relating to the discharge or relating to compliance with the waste discharge requirements;
- c. to inspect monitoring equipment or records; and
- d. to sample any discharge.

2. Reporting Requirements

- a. Pursuant to California Water Code 13267(b), the discharger shall immediately notify the Regional Board by telephone whenever an adverse condition occurred as a result of this discharge; written confirmation shall follow within two weeks. An adverse condition includes, but is not limited to, spills of petroleum products or toxic chemicals, or damage to control facilities that could affect compliance.
- b. Pursuant to California Water Code Section 13260 (c), any proposed material change in the character of the waste, manner or method of treatment or disposal, increase of discharge, or location of discharge, shall be reported to the Regional Board at least 120 days in advance of implementation of any such proposal. This shall include, but not be limited to, all significant soil disturbances.
- c. The owner(s) of, and discharger upon, property subject to waste discharge requirements shall be considered to have a continuing responsibility for ensuring compliance with applicable waste discharge requirements in the operations or use of the owned property. Pursuant to California Water Code Section 13260(c), any change in the ownership and/or operation of property subject to the waste discharge requirements shall be reported to the Regional Board. Notification of applicable waste discharge requirements shall be furnished in writing to the new owners and/or operators and a copy of such notification shall be sent to the Regional Board.
- d. If a discharger becomes aware that any information submitted to the Regional Board is incorrect, the discharger shall immediately notify the Regional Board, in writing, and correct that information.

- e. Reports required by the waste discharge requirements, and other information requested by the Regional Board, must be signed by a duly authorized representative of the discharger. Under Section 13268 of the California Water Code, any person failing or refusing to furnish technical or monitoring reports, or falsifying any information provided therein, is guilty of a misdemeanor and may be liable civilly in an amount of up to one thousand dollars (\$1000) for each day of violation.
- f. If the discharger becomes aware that their waste discharge requirements are no longer needed (because the project will not be built or the discharge will cease) the discharger shall notify the Regional Board in writing and request that their waste discharge requirements be rescinded.

3. Right to Revise Waste Discharge Requirements

The Board reserves the privilege of changing all or any portion of the waste discharge requirements upon legal notice to and after opportunity to be heard is given to all concerned parties.

4. Duty to Comply

Failure to comply with the waste discharge requirements may constitute a violation of the California Water Code and is grounds for enforcement action or for permit termination, revocation and reissuance, or modification.

5. Duty to Mitigate

The discharger shall take all reasonable steps to minimize or prevent any discharge in violation of the waste discharge requirements which has a reasonable likelihood of adversely affecting human health or the environment.

6. Proper Operation and Maintenance

The discharger shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) that are installed or used by the discharger to achieve compliance with the waste discharge requirements. Proper operation and maintenance includes adequate laboratory control, where appropriate, and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by the discharger, when necessary to achieve compliance with the conditions of the waste discharge requirements.

7. Waste Discharge Requirement Actions

The waste discharge requirements may be modified, revoked and reissued, or terminated for cause. The filing of a request by the discharger for waste discharge requirement

modification, revocation and reissuance, termination, or a notification of planned changes or anticipated noncompliance, does not stay any of the waste discharge requirements conditions.

8. Property Rights

The waste discharge requirements do not convey any property rights of any sort, or any exclusive privileges, nor does it authorize any injury to private property or any invasion of personal rights, nor any infringement of federal, state or local laws or regulations.

9. Enforcement

The California Water Code provides for civil liability and criminal penalties for violations or threatened violations of the waste discharge requirements including imposition of civil liability or referral to the Attorney General.

10. Availability

A copy of the waste discharge requirements shall kept and maintained by the discharger and be available at all times to operating personnel.

11. Severability

Provisions of the waste discharge requirements are severable. If any provision of the requirements is found invalid, the remainder of the requirements shall not be affected.

12. Public Access

General public access shall be effectively excluded from treatment and disposal facilities.

13. Transfers

Providing there is no material change in the operation of the facility, this Order may be transferred to a new owner or operation. The owner/operator must request the transfer in writing and receive written approval from the Regional Board Executive Officer.

14. Definitions

- a. "Surface waters" as used in this Order, include, but are not limited to, live streams, either perennial or ephemeral, which flow in natural or artificial water courses and natural lakes and artificial impoundments of waters. "Surface waters" does not include artificial water courses or impoundments used exclusively for wastewater disposal.
- b. "Ground waters" as used in this Order, include, but are not limited to, all subsurface waters being above atmospheric pressure and the capillary fringe of these waters.

15. Storm Protection

All facilities used for collection, transport, treatment, storage, or disposal of waste shall be adequately protected against overflow, washout, inundation, structural damage or a significant reduction in efficiency resulting from a storm or flood having a recurrence interval of once in 100 years.

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
LAHONTAN REGION

**MONITORING AND REPORTING PROGRAM NO. 2002-0030
FOR**

**TAHOE-TRUCKEE SANITATION AGENCY
MARTIS VALLEY WASTEWATER TREATMENT PLANT AND
ASSOCIATED MAINTENANCE PROJECTS**

Nevada County_____

This monitoring and reporting program includes five areas of monitoring:

1. Water rights monitoring
2. Collection system flow monitoring
3. Plant influent and effluent monitoring
4. Receiving waters monitoring; both ground and surface waters
5. Maintenance projects monitoring.

This program shall take effect immediately.

WATER RIGHTS MONITORING

MONITORING

The Tahoe-Truckee Sanitation Agency (TTSA) shall provide annual reports on the total monthly water use within the water service district boundaries of member entities in the Lake Tahoe Basin for the prior calendar year. These reports shall include the following information on a monthly basis:

1. Total water diversion for use (MG)
2. Number and Type of water users served by each water system or subsystem
3. Unit water use rates (gpd)

These reports shall include all water use within the member entities water service areas, for purposes of municipal use, domestic use, agricultural use, irrigation use, and industrial use, excepting use on federal and state owned lands. The data provided in these reports shall be based on direct measurements to the greatest extent practicable, but may rely upon estimating techniques such as those employed in the State Water Resources Control Board's "Report on Water Use and Water Rights, Lake Tahoe Basin" or other similar methods.

REPORTING

The Discharger shall submit annual monitoring reports not later than June 15th for the previous calendar year. The first such report shall be due not later than June 15, 2002. In reporting the data, the Discharger shall arrange the data such that the diversion, period of diversion, amounts, numbers of users, and rate of use are readily discernible.

FLOW MONITORING WITHIN COLLECTION SYSTEM

FLOW MEASUREMENT

Flow monitoring of member districts shall be initiated immediately upon connection of each district to regional system. Flow meters capable of accurately measuring flow shall be installed at all points where an individual district discharges to the regional interceptor.

Flow shall be monitored for:

1. Total daily flow (MG)
2. Daily peak flow rate (MGD)

The Truckee River Canyon is exempt for individual flow monitoring requirements.

CALIBRATION

Each meter shall be calibrated semi-annually under the supervision of a registered civil engineer and the report of the calibration shall be prepared by him/her and submitted within 15 days after calibration.

REPORTING

The Discharger shall submit monthly monitoring reports no later than the 15th day following each monthly monitoring period. The Discharger shall: (1) compute a running seven-day average flow, (2) identify the maximum daily flow and its date, and (3) identify the peak flow rate and its date. In reporting the data, the Discharger shall arrange the data such that the subject district, date, and flow are readily discernible.

TREATMENT PLANT MONITORING

Treatment plant monitoring will consist of measurement of influent flow and composition and detailed analyses of effluent quality.

FLOW MONITORING

A flow meter capable of accurately measuring influent shall be installed downstream of all significant wastewater contributors and above the first unit operation of the treatment plant. The calibration requirements noted in "Collection System Flow Monitoring" above shall apply to this meter also.. Additional accurate flow meters shall be installed as appropriate to enable flow measurement within the treatment plant. The following flows shall be monitored:

1. Total daily influent flow (MG)
2. Peak daily influent flow rate (MGD)
3. Total daily flow of effluent to disposal site (MG)
4. Total daily flow to emergency retention basin (MG)

5. Total daily flow from emergency retention basin to the treatment works (MG)
6. Total daily flow (MG) to and from the emergency storage facilities.

INFLUENT MONITORING

Influent samples shall be collected at the headworks of the plant prior to any treatment process. The following shall constitute the program for monitoring of influent water quality:

Parameter	Units	Sample Type	Frequency
Influent COD	mg/l	24-hour composite	2/week ^{1/}
Influent Total Suspended Solids	mg/l	24-hour composite	2/week ^{1/}
Influent BOD ₅	mg/l	24-hour composite	Weekly ^{2/}
Influent Total Nitrogen	mg/l	24-hour composite	Weekly ^{2/}
Influent Total Phosphorus	mg/l	24-hour composite	Weekly

EFFLUENT MONITORING

Effluent samples shall be collected at the effluent sampler on the effluent line. The following shall constitute the program for monitoring of effluent water quality:

Parameter	Units	Sample Type	Frequency
Turbidity	NTU (range of values)	Continuous ^{2,3/}	---
pH	pH Units	Continuous ^{4/}	---
Chlorine Residual	mg/l (range of values)	Continuous ^{4/}	---
Temperature	°C	Grab	Daily
Turbidity	NTU	24-hour composite	Daily
Dissolved Oxygen	mg/l	Grab	Daily
Total Coliform Organisms	MPN/100 ml or MPC/100 ml	Grab	Daily
COD	mg/l	24-hour composite	2/week
Total Organic Carbon	mg/l	24-hour composite	2/week
Total Phosphorus	mg/l-P	24-hour composite	2/week
Nitrate	mg/l	24-hour composite	3/week
Total Kjeldahl-N	mg/l-N	24-hour composite	3/week
Total Suspended Solids	mg/l	24-hour composite	Weekly
Alkalinity	mg/l CaCO ₃	Grab ^{3/}	Weekly
Chlorine Residual	mg/l	Grab	Weekly
Chloride	mg/l	24-hour composite	Weekly
Trihalomethanes	mg/l	24-hour composite	Quarterly
Phenols	mg/l	24-hour composite	Quarterly
Sulfate	mg/l	24-hour composite	Quarterly
Total Dissolved Solids	mg/l	24-hour composite	Quarterly
Sodium	mg/l	24-hour composite	Quarterly
Calcium	mg/l	24-hour composite	Quarterly
Iron	mg/l	24-hour composite	Quarterly

Arsenic	mg/l	24-hour composite	Annually
Barium	mg/l	24-hour composite	Annually
Boron	mg/l	24-hour composite	Annually
Cadmium	mg/l	24-hour composite	Annually
Hexavalent Chromium	mg/l	24-hour composite	Annually
Lead	mg/l	24-hour composite	Annually
Selenium	mg/l	24-hour composite	Annually
Silver	mg/l	24-hour composite	Annually
Copper	mg/l	24-hour composite	Annually
Manganese	mg/l	24-hour composite	Annually
Zinc	mg/l	24-hour composite	Annually
Nickel	mg/l	24-hour composite	Annually
Strontium	mg/l	24-hour composite	Annually
Magnesium	mg/l	24-hour composite	Annually

^{1/} Every Sunday and Wednesday

^{2/} Alternating Sunday and Wednesday

^{3/} May be taken at overflow from second stage recarbonation basin

^{4/} Use of continuous recording probe is essential

^{5/} Sample may be taken prior to chlorination

MASS LOADS

The Discharger shall calculate for each calendar year total annual mass loads for the following constituents discharged from the treatment plant:

1. Total Dissolved Solids
2. Chloride
3. Total Phosphorus (as P)
4. Total Nitrogen (as N)
5. Total Nitrogen (as N), mass load for period May 1 through October 31

REPORTING

The Discharger shall submit monthly monitoring reports not later than the 15th day of the following month. Each report shall contain the results of appropriate daily, weekly, quarterly, or annual sampling as noted above. Turbidity and pH shall be reported in terms of a daily range of values. The Discharger shall compute and report the 30-day monthly mean and the monthly maximum value for those parameters listed in Section II.A.2 of Board Order No. R6T-2002-(PROPOSED). In reporting the data, the Discharger shall arrange the parameter name, units, date, measured value and computed value so as to be readily discernible and clearly illustrate compliance.

RECEIVING WATER MONITORING

SURFACE WATERS

Station Code^{7/}

T-1

Description^{8/}

Near Polaris; N614, 240, E2, 527, 680

T-2	Just above old Highway 40 bridge; N619, 620, E2, 531, 800
T-3	Just above California-Nevada State Line
M-1	Below dam and above influence of TTSA discharge
M-2	Just above confluence with Truckee River, N16, 360, E2, 535, 280

^{2/} T=Truckee River Station, M= Martis Creek Station

^{8/} Coordinates based on California Grid

All samples shall be grab samples and shall be taken in accordance with the following schedule for Martis Creek and the Truckee River:

Parameter	Units	Station	Frequency
Temperature	°C	All Stations	2/month
Nitrate	mg/l as N	All Stations	2/month
Total Kjeldahl-N	mg/l as N	All Stations	Monthly
Total Phosphorus	mg/l as P	All Stations	Monthly
Ortho-phosphate	mg/l as P	All Stations	Monthly
Total Coliforms	MPN/100ml or MPC/100ml	All Stations	Monthly ^{9/}
Fecal Coliforms	MPN/100ml or MPC/100ml	All Stations	Monthly
Total Iron	mg/l	All Stations	Monthly
Dissolved Oxygen	mg/l	All Stations	Monthly
Alkalinity	mg/l CaCO ₃	All Stations	Monthly
pH	pH units	All Stations	Monthly
Dissolved Organic Carbon	mg/l	All Stations	Monthly
Chloride	mg/l	All Stations	Monthly
Total Dissolved Solids		All Stations	Monthly
Un-Ionized Ammonia	mg/l as N	All Stations	Quarterly
Trihalomethanes	mg/l	T-2, T-3, M-2	Quarterly
Periphyton	gr.dry wt./m ² and gr. Ash free dry wt./m ²	All Stations	2/month (May-Oct)
Periphyton	percent composition ^{10/}	All Stations	2/month (May-Oct)
Benthic Invertebrates ^{11/}	numbers	All Stations	Monthly (June-Oct)

^{9/} 2/week during any period when emergency storage facilities contain sewage

^{10/} Relative percentages (from cell counts) of algae in major groups [Chlorophyta, Chrysophyta
(separate diatoms from other Chrysophyta), Cyanophyta]

^{11/} Invertebrates to be identified to phylum level (insects to order level). Number of individuals/m² in each group to be reported. Insect diversity to be computed based on numbers of individuals in each order.

GROUND WATERS

FINAL MONITORING AT WELL MG-5-TO

Samples of ground water containing treated effluent shall be collected at monitoring well MG-5-TO (Well 31). The following shall constitute the program for monitoring of ground water containing treated effluent at Well MG-5-TO:

Parameter	Units	Sample Type	Frequency
Static Water Level	feet MSL	---	Weekly
COD	mg/l	grab	Weekly
Total Organic Carbon	mg/l	grab	Weekly
Nitrate Nitrogen	mg/l as N	grab	Weekly
Total Kjeldahl-N	mg/l as N	grab	Weekly
Un-ionized Ammonia	mg/l as N	grab	Weekly
Total Phosphorus	mg/l as P	grab	Weekly
Total Fecal Coliform	MPN/100 ml	grab	Weekly
Chlorine Residual	mg/l	grab	Weekly
Chloride	mg/l	grab	Weekly
pH	pH units	grab	Weekly
Alkalinity	mg/l as CaCO ₃	grab	Weekly
Temperature	o C	grab	Weekly
Total Dissolved Solids	mg/l	grab	Weekly
Trihalomethanes	mg/l	grab	Quarterly
Purgeable Halocarbons ^{14/}	ug/l	grab	Annually
Purgeable Aromatics ^{15/}	ug/l	grab	Annually

ADDITIONAL GROUND WATER MONITORING

The following additional ground water monitoring sampling stations are to be maintained:

Station Code ^{12/}	Location ^{13/}
MG-1-TO	East edge of disposal area 17N/17E-7R1M (Well 20)
MG-1-TF	East edge of disposal area 17N/17E-7R1M (Well 1)
MG-2-TO	Martis Valley near Martis Creek, 17N/17E-7J1M (Toups Well)
MG-2-TF	Martis Valley near Martis Creek, 17N/17E-7J1M (Well 23)
MG-4-TO	Martis Valley near Martis Creek, 17N/17E-8F1 (Well 36)
MG-5-TO	Martis Valley near Truckee River, 17N/17ESN1 (Well 31)
MG-6-TO	Martis Valley near Truckee River, 17N/17E (Well 25)
MG-6-TF	Martis Valley near Truckee River, 17N/17E (Well 26)
MG-7-TO	Martis Valley near Martis Creek, 17N/17E (Well 34)
Upgradient	A specific upgradient well site (Well 24)

^{12/}MG=Martis Valley ground water body; TO= Tahoe Outwash; TF=Truckee Formation

^{13/}Well Location System, U.S. Geological Survey

For those monitoring wells with suffix "TO", the casing shall only extend to a depth at which the top of the Truckee Formation is encountered and shall be perforated to within 20 feet of the ground surface.

For those monitoring wells with the suffix "TF", the casing shall extend to at least 20 feet below the first clay layer encountered below the Tahoe Outwash and shall be sealed above this depth and perforated below. Exact casing and perforation depths shall be determined in the field by a registered civil engineer or a certified engineering geologist. Well construction shall conform to applicable ordinances of the County of Nevada and *Water Well Standards for the State of California* (DWR Bulletin No. 74).

Sampling of the wells shall be conducted by drawing the appropriate sample volume from the upper 3 feet of ground water encountered in each well.

All samples shall be grab samples and shall be drawn according to the following schedules:

Sampling of Stations MG-1-TO, MG-2-TO, MG-4-TO, MG-6-TO AND MG-7-TO

<u>Parameter</u>	<u>Units</u>	<u>Frequency</u>
Static Water Level	feet MSL	Monthly
Nitrate Nitrogen	mg/l as N	Monthly
Total Kjeldahl Nitrogen	mg/l as N	Monthly
Total Phosphorus	mg/l as P	Monthly
Total Organic Carbon	mg/l	Monthly
pH	pH units	Monthly
Temperature	°C	Monthly
Chloride	mg/l	Monthly
Total Dissolved Solids	mg/l	Monthly
Alkalinity	mg/l as CaCO ₃	Quarterly
Trihalomethanes	mg/l	Quarterly
Un-Ionized Ammonia	mg/l as N	Quarterly
Total Fecal Coliforms	MPN/100ml or MFC/100ml	Semi-annually (Sta. MG-1-TO and MG-2-TO)
Purgeable Halocarbons ^{14/}	ug/l	Annually
Purgeable Aromatics ^{15/}	ug/l	Annually

^{14/}EPA method 601 for samples from wells MG-1-TO, MG-2-TO and the designated upgradient well

^{15/}EPA method 602 plus xylene for samples from wells MG-1-TP, MT-2-TO and the designated upgradient well

Sampling of Stations MG-1-TF, MG-2-TF and MG-6-TF

<u>Parameter</u>	<u>Units</u>	<u>Frequency</u>
Static Water Level	feet MSL	Monthly
Nitrate	mg/l as N	Semi-annually
Total Organic Carbon	mg/l	Semi-annually
PpH	pH units	Semi-annually

Temperature	°C	Semi-annually
Chloride	mg/l	Semi-annually
Total Dissolved Solids	mg/l	Semi-annually
Trihalomethanes	mg/l	Semi-annually
Total Fecal Coliform	MPN/100ml or MFC/100ml	Semi-annually

REPORTING

Monthly monitoring reports shall be submitted by the Discharger not later than the 15th day of the following month. In reporting the data, the Discharger shall arrange the data such that the station code, date, measured value, and applicable standard are clearly discernible.

MAINTENANCE PROJECTS

POST-CONSTRUCTION PHASE

An inspection of all maintenance project sites shall be made by the Discharger twice each year, about every six months when not covered by snow. The purpose of these inspections is to discover potential erosion and surface runoff problems on project sites so that corrective measures may be immediately undertaken.

Any erosion or surface runoff problems found as a result of these inspections shall be clearly described and the corrective measures proposed by the Discharger shall be included in the monitoring report. In the event that no such problems are found on the subject property, a statement certifying this condition must be included for each semiannual inspection.

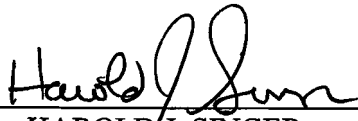
GENERAL

1. The Discharger shall comply with "General Provisions for Monitoring and Reporting", dated September 1, 1994, which is attached to and made a part of this Monitoring and Reporting Program.
2. All analyses shall be performed in accordance with the latest edition of *Standard Methods for the Examination of Water and Wastewater* or the *Manual of Methods for Chemical Analysis for Water and Waste* unless otherwise noted, in a laboratory certified to perform such analyses by the California Department of Health, or approved by the Executive Officer.
3. In monthly monitoring reports, the Discharger shall note and explain any unusual occurrence such as failure to any treatment unit or non-compliance with any waste discharge requirement, effluent limitations or receiving water limitation.
4. The February monitoring report of each year shall include trend analyses for the previous calendar year and a comparison of annual means (mean of monthly means) with annual means from previous years to extend back to pre-discharge. Trend analyses will be provided for all surface water parameters at all surface water stations. The trend analyses for

ground waters will be performed for all ground water parameters at wells MG-2, MG-4 and 5-TO. The trend analysis shall include an assessment of any changes to ground water flow direction and gradients, and a discussion of seasonal, spatial and temporal trends if any. If appropriate to make the information understandable, this report shall include summary data tables, graphs, maps of constituent levels and appended analytical reports. The report shall identify any violation of permit limitations shown by this data. Additionally, trend analysis shall be provided to reflect the changes in monthly and annual mean flows through the plant and from each member entity.

5. All monitoring reports will be reviewed and signed by a registered civil engineer who is routinely responsible for conducting this Monitoring and Reporting Program.
6. A detailed QA/QC program shall be established to include, but not be limited to, duplicate analysis, split sample analysis by an alternative laboratory and an analysis of spike samples. Details and results of the QA/QC program shall be reported annually with the first report due by July 1, 2002.

Ordered by:


HAROLD J. SINGER
EXECUTIVE OFFICER

Dated

May 8, 2002

Attachments: A. General Provisions for Monitoring and Reporting
B. Map of Monitoring Locations

TJP/cgT: TTSA2002PROP.MRP

ATTACHMENT "A"
CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
LAHONTAN REGION

GENERAL PROVISIONS
FOR MONITORING AND REPORTING

1. **SAMPLING AND ANALYSIS**

- a. All analyses shall be performed in accordance with the current edition(s) of the following documents:
 - i. Standard Methods for the Examination of Water and Wastewater
 - ii. Methods for Chemical Analysis of Water and Wastes, EPA
- b. All analyses shall be performed in a laboratory certified to perform such analyses by the California State Department of Health Services or a laboratory approved by the Regional Board Executive Officer. Specific methods of analysis must be identified on each laboratory report.
- c. Any modifications to the above methods to eliminate known interferences shall be reported with the sample results. The methods used shall also be reported. If methods other than EPA-approved methods or Standard Methods are used, the exact methodology must be submitted for review and must be approved by the Regional Board Executive Officer prior to use.
- d. The discharger shall establish chain-of-custody procedures to insure that specific individuals are responsible for sample integrity from commencement of sample collection through delivery to an approved laboratory. Sample collection, storage, and analysis shall be conducted in accordance with an approved Sampling and Analysis Plan (SAP). The most recent version of the approved SAP shall be kept at the facility.
- e. The discharger shall calibrate and perform maintenance procedures on all monitoring instruments and equipment to ensure accuracy of measurements, or shall insure that both activities will be conducted. The calibration of any wastewater flow measuring device shall be recorded and maintained in the permanent log book described in 2.b, below.
- f. A grab sample is defined as an individual sample collected in fewer than 15 minutes.
- g. A composite sample is defined as a combination of no fewer than eight individual samples obtained over the specified sampling period at equal intervals. The volume of each individual sample shall be proportional to the discharge flow rate at the time of sampling. The sampling period shall equal the discharge period, or 24 hours, whichever period is shorter.

2. OPERATIONAL REQUIREMENTS

a. Sample Results

Pursuant to California Water Code Section 13267(b), the discharger shall maintain all sampling and analytical results including: strip charts; date, exact place, and time of sampling; date analyses were performed; sample collector's name; analyst's name; analytical techniques used; and results of all analyses. Such records shall be retained for a minimum of three years. This period of retention shall be extended during the course of any unresolved litigation regarding this discharge, or when requested by the Regional Board.

b. Operational Log

Pursuant to California Water Code Section 13267(b), an operation and maintenance log shall be maintained at the facility. All monitoring and reporting data shall be recorded in a permanent log book.

3. REPORTING

- a. For every item where the requirements are not met, the discharger shall submit a statement of the actions undertaken or proposed which will bring the discharge into full compliance with requirements at the earliest time, and shall submit a timetable for correction.
- b. Pursuant to California Water Code Section 13267(b), all sampling and analytical results shall be made available to the Regional Board upon request. Results shall be retained for a minimum of three years. This period of retention shall be extended during the course of any unresolved litigation regarding this discharge, or when requested by the Regional Board.
- c. The discharger shall provide a brief summary of any operational problems and maintenance activities to the Board with each monitoring report. Any modifications or additions to, or any major maintenance conducted on, or any major problems occurring to the wastewater conveyance system, treatment facilities, or disposal facilities shall be included in this summary.
- d. Monitoring reports shall be signed by:
 - i. In the case of a corporation, by a principal executive officer at least of the level of vice-president or his duly authorized representative, if such representative is responsible for the overall operation of the facility from which the discharge originates;
 - ii. In the case of a partnership, by a general partner;
 - iii. In the case of a sole proprietorship, by the proprietor; or

- iv. In the case of a municipal, state or other public facility, by either a principal executive officer, ranking elected official, or other duly authorized employee.
- e. Monitoring reports are to include the following:
 - i. Name and telephone number of individual who can answer questions about the report.
 - ii. The Monitoring and Reporting Program Number.
 - iii. WDID Number 6A265300900.
- f. Modifications

This Monitoring and Reporting Program may be modified at the discretion of the Regional Board Executive Officer.

4. NONCOMPLIANCE

Under Section 13268 of the Water Code, any person failing or refusing to furnish technical or monitoring reports, or falsifying any information provided therein, is guilty of a misdemeanor and may be liable civilly in an amount of up to one thousand dollars (\$1,000) for each day of violation.

T-3 near
stateline

Legend:

•	Monitoring Well
T, M	Surface Monitoring Station
Station Code	TTSA Well Number
MG-1-TO	20
MG-1-TF	1
MG-2-TO	TOUPS
MG-2-TF	23
MG-4-TO	36
MG-5-TO	31
MG-6-TO	25
MG-6-TF	26
MG-7-TO	34
UpGradient	24
-----	TTSA Property Boundary (approximate)
- - - - -	Surface Waterway

